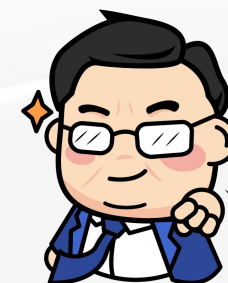


26高数强化 (16)

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主讲 武忠祥 教授



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你就慢了



@武忠祥老师 @考研可爱因子



第二节 偏导数与全微分的计算

本节内容要点

一. 考试内容要点精讲

(一) 复合函数求导法 ✓

(二) 隐函数求导法 ✓



还不关注，
你就慢了



二. 常考题型方法与技巧

题型一 求一点处的偏导数与全微分

题型二 求已给出具体表达式函数的偏导数与全微分

题型三 含有抽象函数的复合函数的偏导数与全微分

题型四 隐函数的偏导数与全微分

一. 考试内容要点精讲

(一) 复合函数求导法

设 $u = u(x, y), v = v(x, y)$ 可导, $z = f(u, v)$ 在相应点有连续一阶偏导数, 则

$$\frac{\partial z}{\partial x} = \frac{\partial f}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial f}{\partial v} \frac{\partial v}{\partial x}$$

$$\frac{\partial z}{\partial y} = \frac{\partial f}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial f}{\partial v} \frac{\partial v}{\partial y}$$

全微分形式不变性

设 $z = f(u, v), u = u(x, y), v = v(x, y)$ 都有连续一阶偏导数

$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$$

$$dz = \frac{\partial z}{\partial u} du + \frac{\partial z}{\partial v} dv$$

$$dy = y'_x dx = y'_u u'_x dx = y'_u du$$
$$y = f(u), u = \varphi(x) \rightarrow y = f[\varphi(x)] \quad y'_x = f'(u) \varphi'(x)$$



(二) 隐函数求导法

1. 由一个方程所确定的隐函数

设 $F(x, y, z)$ 有连续一阶偏导数, $F_z \neq 0$ $z = z(x, y)$ 由 $F(x, y, z) = 0$ 所确定.

方法: (1) 公式 $\frac{\partial z}{\partial x} = -\frac{F_x}{F_z}, \frac{\partial z}{\partial y} = -\frac{F_y}{F_z};$

(2) 等式两边求导

$$F_x + F_z \frac{\partial z}{\partial x} = 0, \quad F_y + F_z \frac{\partial z}{\partial y} = 0,$$

(3) 利用微分形式不变性

$$F_x dx + F_y dy + F_z dz = 0$$

Handwritten notes and diagrams:

- $F'_y = 2y \neq 0, y \neq 0$
- $F'_x \neq 0 \Rightarrow x = x(y, z)$
- $F(x, y) = x^2 + y^2 - 1 = 0$
- $x^2 + y^2 = 1$
- $F'_y \neq 0$
- $y = y(x)$
- Diagram of a circle $x^2 + y^2 = 1$ with a point (x, y) marked on it.

$$dz = \left(\frac{\partial z}{\partial x} \right) dx + \left(\frac{\partial z}{\partial y} \right) dy$$

2. 由方程组所确定的隐函数 (仅数一要求)

设 $u = u(x, y), v = v(x, y)$ 由 $\begin{cases} F(x, y, u, v) = 0 \\ G(x, y, u, v) = 0 \end{cases}$ 所确定.

方法:

(1) 等式两边求导

$$\begin{cases} F_x + F_u \frac{\partial u}{\partial x} + F_v \frac{\partial v}{\partial x} = 0 \\ G_x + G_u \frac{\partial u}{\partial x} + G_v \frac{\partial v}{\partial x} = 0 \end{cases}$$

(2) 微分形式不变性

$$\begin{cases} F_x dx + F_y dy + F_u du + F_v dv = 0 \\ G_x dx + G_y dy + G_u du + G_v dv = 0 \end{cases}$$

$$du = (u_x) dx + (u_y) dy$$

$$dv = (v_x) dx + (v_y) dy$$

题型一 求一点处的偏导数与全微分

【例1】 设 $f(x, y) = \begin{cases} \frac{\sqrt{|x|}}{x^2 + y^2} \sin(x^2 + y^2) & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$

求 $f_x(0, 0)$ 和 $f_y(0, 0)$.

【解】 由于 $\lim_{\Delta x \rightarrow 0} \frac{f(\Delta x, 0) - f(0, 0)}{\Delta x}$

$$= \lim_{\Delta x \rightarrow 0} \frac{\frac{\sqrt{|\Delta x|}}{(\Delta x)^2} \sin(\Delta x)^2}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\sqrt{|\Delta x|}}{\Delta x} = \infty.$$

则 $f_x(0, 0)$ 不存在；而

$$f_y(0, 0) = \lim_{\Delta y \rightarrow 0} \frac{f(0, \Delta y) - f(0, 0)}{\Delta y} = \lim_{\Delta y \rightarrow 0} \frac{0 - 0}{\Delta y} = 0.$$

先线后求

$$f(x, 0) = \begin{cases} \frac{\sqrt{|x|} \sin x^2}{x^2}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

$$\lim_{x \rightarrow 0} \frac{\frac{\sqrt{|x|}}{x^2} \sin x^2}{x} = \infty$$

$$f(0, y) = \begin{cases} 0, & y \neq 0 \\ 0, & y = 0 \end{cases} \quad f'_y(0, 0) = 0$$

【例2】 设 $f(x, y) = \frac{2x + 3y}{1 + xy\sqrt{x^2 + y^2}}$, 求 $f_x(0,0)$ 和 $f_y(0,0)$.

【解】 $f_x(0,0) = \frac{d}{dx} f(x,0)|_{x=0} = \frac{d}{dx} (2x)|_{x=0} = 2;$

$f_y(0,0) = \frac{d}{dy} f(0,y)|_{y=0} = \frac{d}{dy} (3y)|_{y=0} = 3;$

① $f'_x(x,y) \cdot f'_y(x,y)$

② 或 $x=0, y \neq 0$

“先代后求”

【例3】设 $z = \ln(1 + xy^2)$ ，则 $\frac{\partial^2 z}{\partial x \partial y} \Big|_{(0,1)} = \underline{\hspace{2cm}}$.

【解】 $\frac{\partial z}{\partial x} = \frac{y^2}{1 + xy^2}$

$x=0$

y

$$\frac{\partial^2 z}{\partial x \partial y} \Big|_{(0,1)} = \frac{d}{dy} \left(\frac{\partial z(0, y)}{\partial x} \right) \Big|_{y=1} = \frac{d}{dy} (y^2) \Big|_{y=1} = 2$$

【例4】设 $f(x, y, z) = z\sqrt{\frac{x}{y}}$, 则 $df(1, 1, 1) = \underline{\hspace{2cm}}$.

【解】 $\underline{f_x(1, 1, 1)} = \frac{d}{dx} f(x, \underline{1}, \underline{1}) \Big|_{x=1} = \frac{d}{dx} (\underline{x}) \Big|_{x=1} = 1$

$$\underline{f_y(1, 1, 1)} = \frac{d}{dy} f(1, \underline{y}, \underline{1}) \Big|_{y=1} = \frac{d}{dy} \left(\frac{1}{\underline{y}} \right) \Big|_{y=1} = \underline{-\frac{1}{y^2}} \Big|_{y=1} = \underline{-1}$$

$$\underline{f_z(1, 1, 1)} = \frac{d}{dz} f(\underline{1}, \underline{1}, \underline{z}) \Big|_{z=1} = \frac{d}{dz} (\underline{1}) \Big|_{z=1} = \underline{0}$$

故 $df(1, 1, 1) = dx - dy$

题型二 求已给出具体表达式函数的偏导数与全微分

【例1】设 $z = (x^2 + y^2)e^{-\arctan \frac{y}{x}}$, 求 $\frac{\partial z}{\partial x}$, $\frac{\partial z}{\partial y}$ 及 dz .

【例2】设 $z = (1 + x^2 + y^2)^{xy}$ ，求 $\frac{\partial z}{\partial x}$ 及 $\frac{\partial z}{\partial y}$ 。

【解1】 $z = e^{xy \ln(1+x^2+y^2)}$

【解2】 $\ln z = xy \ln(1+x^2+y^2)$

【解3】 令 $u = 1 + x^2 + y^2$, $v = xy$, 则 $z = u^v$

$$\begin{aligned} \frac{\partial z}{\partial x} &= \frac{\partial z}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial x} = v u^{v-1} 2x + u^v \ln u \cdot y \\ &= (1+x^2+y^2)^{xy} \left[\frac{2x^2 y}{1+x^2+y^2} + y \ln(1+x^2+y^2) \right] \end{aligned}$$

同理可得 $\frac{\partial z}{\partial y}$ 。

$(x^a)^b = x^{ab}$
 $(x^a)^b = a x^{a-1} \cdot b x^{b-1} + (1+x^2)^{\ln(1+x^2+y^2)} \cdot xy$

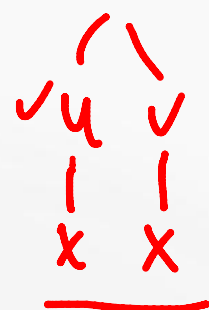
$(a^x)' = a^x \ln a$



$y = (1+x^2)^{xy}$

$u = 1+x^2, v = xy$

$y = u^v$



$y'_x = v u^{v-1} 2x + u^v \ln u \cdot y'_x$

【例3】若函数 $z = f(x, y)$ 满足 $\frac{\partial^2 z}{\partial y^2} = 2$, 且 $f(x, 1) = x + 2$, 又

$f_y(x, 1) = x + 1$, 则 $f(x, y)$ 等于

- ~~A) $y^2 + (x-1)y - 2$;~~
~~B) $y^2 + (x+1)y + 2$;~~
~~C) $y^2 + (x-1)y + 2$;~~
~~D) $y^2 + (x+1)y - 2$;~~

【解1】容易验证 (C) 正确.

【解2】由 $\frac{\partial^2 z}{\partial y^2} = 2$ 知 $\frac{\partial z}{\partial y} = \int 2dy = 2y + \varphi(x)$

由 $f_y(x, 1) = 1 + x$ 知

$$1 + x = 2 + \varphi(x) \Rightarrow \varphi(x) = x - 1 \Rightarrow \frac{\partial z}{\partial y} = 2y + x - 1$$

$$z = \int (2y + x - 1)dy = y^2 + y(x - 1) + \psi(x)$$

由 $f(x, 1) = x + 2$ 知

$$x + 2 = 1 + (x - 1) + \psi(x) \Rightarrow \psi(x) = 2$$

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$$f'(x) = 2$$

$$f(x, 1)$$

【例4】已知 $\frac{\partial^2 z}{\partial x \partial y} = 1$ ，且当 $x=0$ 时， $z = \sin y$ ；当 $y=0$ 时，

$z = \sin x$ 则 $z(x, y) =$

【解1】由 $\frac{\partial^2 z}{\partial x \partial y} = 1$ 知， $\frac{\partial z}{\partial x} = \int 1 dy = y + \varphi(x)$

$$z = \int [y + \varphi(x)] dx = xy + \int \varphi(x) dx + \psi(y) = xy + g(x) + \psi(y)$$

由 $x=0$ 时， $z = \sin y$ 知， $\sin y = g(0) + \psi(y)$

由 $y=0$ 时， $z = \sin x$ 知， $\sin x = g(x) + \psi(0)$

$$z = xy + \sin x + \sin y - g(0) - \psi(0)$$

$$g(0) + \psi(0) = 0$$

故 $z(x, y) = xy + \sin x + \sin y$

$$z(x, 0) = \sin x$$

【解2】

$$z'_x(x, 0) = \cos x$$

$$\varphi(x) = \cos x$$

$$0 = g(0) + \psi(0)$$

$$\frac{\partial z}{\partial x} = y + \cos x$$

$$z = \int (y + \cos x) dx$$

$$= xy + \sin x + \psi(y)$$

$$\sin y = \psi(y)$$

【例5】已知 $(axy^3 - y^2 \cos x)dx + (1 + by \sin x + 3x^2 y^2)dy = d\zeta(x, y)$

是某一函数的全微分, 则 a, b 取值分别为:

- (A) -2 和 2; (B) 2 和 -2; (C) -3 和 3; (D) 3 和 -3;

【解】 $df(x, y) = (axy^3 - y^2 \cos x)dx + (1 + by \sin x + 3x^2 y^2)dy$

则 $\frac{\partial f}{\partial x} = axy^3 - y^2 \cos x, \quad \frac{\partial f}{\partial y} = 1 + by \sin x + 3x^2 y^2$

从而有 $\frac{\partial^2 f}{\partial x \partial y} = 3axy^2 - 2y \cos x, \quad \frac{\partial^2 f}{\partial y \partial x} = by \cos x + 6xy^2$

由于 $\frac{\partial^2 f}{\partial x \partial y}$ 和 $\frac{\partial^2 f}{\partial y \partial x}$ 都连续, 从而有 $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$ 即

$3axy^2 - 2y \cos x = by \cos x + 6xy^2$

则 $\begin{cases} 3a = 6, \\ b = -2, \end{cases}$ 即 $\begin{cases} a = 2 \\ b = -2 \end{cases}$, 故应选 (B).

【注】若 $P(x, y)$ 和 $Q(x, y)$ 有一阶连续偏导数, 且

$P(x, y)dx + Q(x, y)dy$ 是某一函数的全微分, 则

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$$

$$Pdx + Qdy = df(x, y)$$

$$P = \frac{\partial f}{\partial x}, \quad Q = \frac{\partial f}{\partial y}$$
$$\frac{\partial P}{\partial y} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial Q}{\partial x} = \frac{\partial^2 f}{\partial y \partial x}$$

【例6】设 $f(x)$ 有连续一阶导数，

$$(xy - yf(x))dx + (f(x) + y^2)dy = du(x, y),$$

求 $f(x)$ 及 $u(x, y)$ ，其中 $f(0) = -1$ 。

【解】由题设知 $x - f(x) = f'(x)$

$$f'(x) + f(x) = x$$

$$y d(x-1) + (x-1)dy = d(y(x-1))$$

$$f(x) = (x-1) + Ce^{-x}$$

由 $f(0) = -1$ 知， $C = 0$ ， $f(x) = x - 1$

$$du(x, y) = ydx + [(x-1) + y^2]dy$$

$$u(x, y) = y(x-1) + \frac{1}{3}y^3 + C$$

$$u = y(x-1) + \frac{1}{3}y^3 + C$$

$$\textcircled{1} \frac{du+dv}{u+dv} = d(u+v) \quad \frac{vdu-udv}{v^2} = d\left(\frac{u}{v}\right)$$

$$3:10 \quad \textcircled{1} \frac{du+dv}{u+dv} = d(u+v)$$

$$\left(\frac{\partial u}{\partial x}\right) = y \Rightarrow u = \int y dx = xy + \varphi(y)$$

$$\frac{\partial u}{\partial y} = x + \varphi'(y)$$

$$\varphi'(y) = -1 + y^2$$

$$\varphi(y) = -y + \frac{1}{3}y^3 + C$$

$$du(x, y) = [y d(x-1) + (x-1)dy] + d\frac{1}{3}y^3$$

$$= d(y(x-1)) + d\frac{1}{3}y^3 = d\left[y(x-1) + \frac{1}{3}y^3\right]$$

题型三 含有抽象函数的复合函数 偏导数与全微分

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【例1】设函数 $f(x, y)$ 可微且, $f(x+1, e^x) = x(x+1)^2$, $f(x, x^2) = 2x^2 \ln x$, 则 $df(1, 1) =$

(A) $dx + dy$ (B) $dx - dy$ (C) dy (D) $-dy$

【解1】直接法 ① $f'_1(x+1, e^x) + e^x f'_2(x+1, e^x) = (x+1)^2 + 2x(x+1)$

② $f'_1(x, x^2) + 2xf'_2(x, x^2) = 4x \ln x + 2x$

$$f'_1(1, 1) + f'_2(1, 1) = 1 \quad f'_1(1, 1) + 2f'_2(1, 1) = 2$$

$$f'_1(1, 1) = 0 \quad f'_2(1, 1) = 1$$

【解2】排除法 $f(x, y) = x^2 \ln y$

$$f_x(1, 1) = 0, f_y(1, 1) = 1$$

$$df(1, 1) = dy$$

【例3】设 $z = f(xy, x^2 + y^2)$, 求

$\frac{\partial z}{\partial x}, \frac{\partial^2 z}{\partial x \partial y}$

其中 $f(u, v)$ 有二阶

连续偏导数.

$f_1(xy, x^2 + y^2)$

【解】 $\frac{\partial z}{\partial x} = y f_1 + 2x f_2$

$$\frac{\partial^2 z}{\partial x \partial y} = f_1 + y[x f_{11} + 2y f_{12}] + 2x[f_{21}x + f_{22} \cdot 2y]$$

$$= f_1 + xy[f_{11} + 4f_{22}] + 2(x^2 + y^2)f_{12}$$

$f_1(xy, x^2 + y^2)$

$f_1(x, y)$

$$\frac{\partial^2 z}{\partial x \partial y} = f_1 + y f_{11} x + y f_{12} 2y + 2x f_{22} 2y + 2x f_{21} x$$

? X

【例4】设 $f(x, y)$ 可微, 又 $f(0, 0) = 0$, $f'_x(0, 0) = a$, $f'_y(0, 0) = b$ 且

$$g(t) = f[t, f(t, t^2)], \text{ 求 } g'(0).$$

【解】 $g'(t) = \underbrace{f'_1(t, f(t, t^2))}_{f'_1(0,0)} + \underbrace{f'_2(t, f(t, t^2))}_{f'_2(0,0)} \cdot [f'_1(t, t^2) + f'_2(t, t^2) \cdot 2t]$

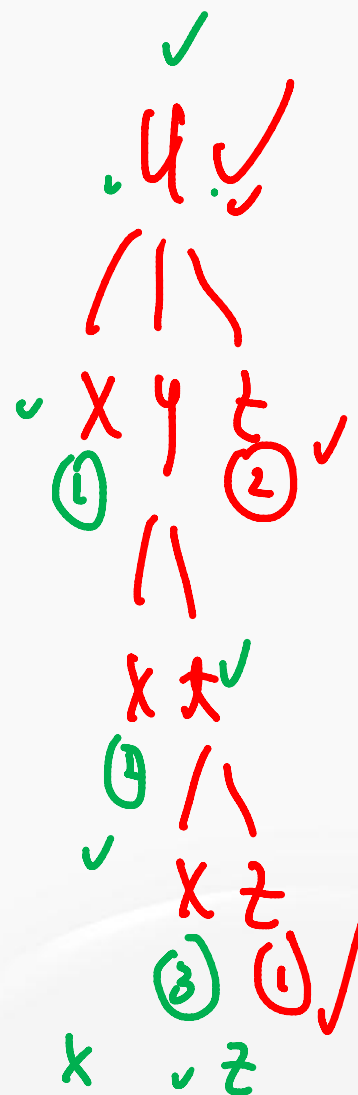
$$g'(0) = \underbrace{a}_{f'_1(0,0)} + \underbrace{b}_{f'_2(0,0)} [a + 0 \times b] = a(1 + b)$$

【例5】设 $u = f(x, y, z)$, $y = \varphi(x, t)$, $t = \psi(x, z)$, 其中 f, φ, ψ

可微, 求 $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial z}$.

【解】
$$\frac{\partial u}{\partial x} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \frac{\partial \varphi}{\partial x} + \frac{\partial f}{\partial y} \frac{\partial \varphi}{\partial t} \frac{\partial \psi}{\partial x}$$

$$\frac{\partial u}{\partial z} = \frac{\partial f}{\partial y} \frac{\partial \varphi}{\partial t} \frac{\partial \psi}{\partial z} + \frac{\partial f}{\partial z}$$



【例7】设函数 $u = f(x, y)$ 具有二阶连续偏导数，且满足 $4\frac{\partial^2 u}{\partial x^2} + 12\frac{\partial^2 u}{\partial x\partial y} + 5\frac{\partial^2 u}{\partial y^2} = 0$ 。
 确定 a, b 的值，使等式在变换 $\xi = x + ay, \eta = x + by$ 下简化为 $\frac{\partial^2 u}{\partial \xi \partial \eta} = 0$ 。

【解1】

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial \xi} + \frac{\partial u}{\partial \eta}$$

$$\frac{\partial u}{\partial y} = a \frac{\partial u}{\partial \xi} + b \frac{\partial u}{\partial \eta}$$

$$\frac{\partial^2 u}{\partial x \partial y} = a \frac{\partial^2 u}{\partial \xi^2} + (a+b) \frac{\partial^2 u}{\partial \xi \partial \eta} + b \frac{\partial^2 u}{\partial \eta^2}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial \xi^2} + 2 \frac{\partial^2 u}{\partial \xi \partial \eta} + \frac{\partial^2 u}{\partial \eta^2}$$

$$\frac{\partial^2 u}{\partial y^2} = a^2 \frac{\partial^2 u}{\partial \xi^2} + 2ab \frac{\partial^2 u}{\partial \xi \partial \eta} + b^2 \frac{\partial^2 u}{\partial \eta^2}$$

$$(5a^2 + 12a + 4) \frac{\partial^2 u}{\partial \xi^2} + [10ab + 12(a+b) + 8] \frac{\partial^2 u}{\partial \xi \partial \eta} + (5b^2 + 12b + 4) \frac{\partial^2 u}{\partial \eta^2} = 0$$

$$\begin{cases} 5a^2 + 12a + 4 = 0 \\ 5b^2 + 12b + 4 = 0 \end{cases}$$

$$10ab + 12(a+b) + 8 \neq 0$$

【例7】设函数 $u = f(x, y)$ 具有二阶连续偏导数，且满足 $4 \frac{\partial^2 u}{\partial x^2} + 12 \frac{\partial^2 u}{\partial x \partial y} + 5 \frac{\partial^2 u}{\partial y^2} = 0$.

确定 a, b 的值，使等式在变换 $\xi = x + ay, \eta = x + by$ 下简化为 $\frac{\partial^2 u}{\partial \xi \partial \eta} = 0$.

【解2】由 $\xi = x + ay, \eta = x + by$ 解得

$$\begin{cases} x = \frac{a\eta - b\xi}{a-b}, \\ y = \frac{\xi - \eta}{a-b}, \end{cases}$$

$$\frac{\partial u}{\partial \xi} = \frac{-b}{a-b} \frac{\partial u}{\partial x} + \frac{1}{a-b} \frac{\partial u}{\partial y}$$

$$\frac{\partial^2 u}{\partial \xi \partial \eta} = \frac{-ab}{(a-b)^2} \frac{\partial^2 u}{\partial x^2} + \frac{a+b}{(a-b)^2} \frac{\partial^2 u}{\partial x \partial y} + \frac{-1}{(a-b)^2} \frac{\partial^2 u}{\partial y^2}$$

欲使 $\frac{\partial^2 u}{\partial \xi \partial \eta} = 0$ ，即 $-ab \frac{\partial^2 u}{\partial x^2} + (a+b) \frac{\partial^2 u}{\partial x \partial y} - \frac{\partial^2 u}{\partial y^2} = 0$

$$\frac{-ab}{4} = \frac{a+b}{12} = \frac{-1}{5}$$

【例8】设 $f(u)$ 具有二阶连续导数，而 $z = f(e^x \sin y)$ 满足方程

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = ze^{2x} \text{ 求 } f(x).$$

【解】令 $u = e^x \sin y$ ，则

$$\frac{\partial z}{\partial x} = f'(u)e^x \sin y,$$

$$\frac{\partial z}{\partial y} = f'(u)e^x \cos y,$$

$$\frac{\partial^2 z}{\partial x^2} = f''(u)e^{2x} \sin^2 y + f'(u)e^x \sin y$$

$$\frac{\partial^2 z}{\partial y^2} = f''(u)e^{2x} \cos^2 y - f'(u)e^x \sin y$$

$$f''(u) = f(u)$$

$$f''(u) - f(u) = 0$$

$$f(u) = c_1 e^u + c_2 e^{-u}$$

$$f''(u)e^{2x} = f(u)e^{2x}$$

$$r^2 - 1 = 0$$

【例9】设 (r, θ) 为极坐标, $u = u(r, \theta)$ 具有二阶连续偏导数,

并满足 $\frac{\partial u}{\partial \theta} \equiv 0$ 且 $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$, 求 $u(r, \theta)$.

【解】由 $\frac{\partial u}{\partial \theta} = 0$ 知, u 仅为 r 的函数, 令 $u = \varphi(r), r = \sqrt{x^2 + y^2}$

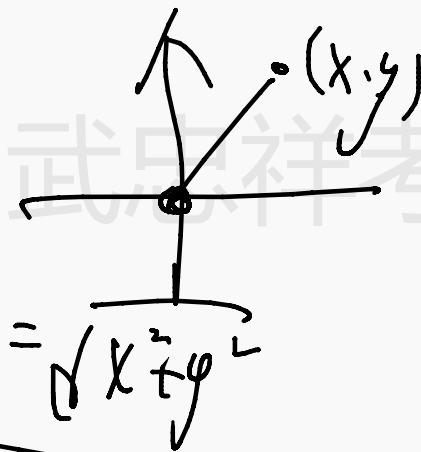
$$\frac{\partial u}{\partial x} = \varphi'(r) \frac{x}{\sqrt{x^2 + y^2}} = \varphi'(r) \frac{x}{r}$$

$$\frac{\partial^2 u}{\partial x^2} = \varphi''(r) \frac{x^2}{r^2} + \varphi'(r) \frac{r - \frac{x^2}{r}}{r^2} = \varphi''(r) \frac{x^2}{r^2} + \varphi'(r) \left(\frac{1}{r} - \frac{x^2}{r^3} \right)$$

$$\frac{\partial^2 u}{\partial y^2} = \varphi''(r) \frac{y^2}{r^2} + \varphi'(r) \left(\frac{1}{r} - \frac{y^2}{r^3} \right)$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \varphi''(r) + \varphi'(r) \frac{1}{r} = 0$$

$$\varphi(r) = C_1 \ln r + C_2 \quad (r\varphi')' = 0$$



$$r\varphi'(r) = C_1$$

$$\varphi'(r) = \frac{C_1}{r}$$

$$\varphi(r) = C_1 \ln r + C_2$$

$$* \quad r\varphi'' + \varphi' = 0$$

$$f(x, y) = x^2 + y^2, \quad f(tx, ty) = t^2 f(x, y)$$

$$y' = f\left(\frac{y}{x}\right)$$

【例10】若对任意 $t > 0$ 有 $f(tx, ty) = t^n f(x, y)$ ，则称函数 $f(x, y)$

是 n 次齐次函数，试证：若 $f(x, y)$ 可微，则 $f(x, y)$ 是

$$n \text{ 次齐次函数} \Leftrightarrow x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = n f(x, y)$$

$$f\left(\frac{tx}{tx}, \frac{ty}{tx}\right) = t \cdot f\left(\frac{y}{x}\right)$$

【证】必要性 由 $f(tx, ty) = t^n f(x, y)$ 得

$$x f_1(tx, ty) + y f_2(tx, ty) = n t^{n-1} f(x, y)$$

$$\text{令 } t = 1 \text{ 得 } x f_1(x, y) + y f_2(x, y) = n f(x, y)$$

$$\text{即 } x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = n f(x, y)$$

充分性 令 $F(t) = f(tx, ty) (t > 0)$ 则 $\frac{dF}{dt} = x f_1(tx, ty) + y f_2(tx, ty)$

$$t \frac{dF}{dt} = [tx] f_1(tx, ty) + [ty] f_2(tx, ty) = n f(tx, ty) = n F(t)$$

$$\frac{dF}{F} = \frac{n}{t} dt, \quad F(t) = C t^n, \quad F(1) = C, \quad F(1) = f(x, y)$$

$$F(t) = t^n f(x, y)$$

【例】设 $z = xyf(\frac{y}{x})$, 且 $f(u)$ 可导, 若 $x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = xy(\ln y - \ln x)$, 则 ()

(A) $f(1) = \frac{1}{2}, f'(1) = 0.$ (B) $f(1) = 0, f'(1) = \frac{1}{2}.$

(C) $f(1) = \frac{1}{2}, f'(1) = 1.$ (D) $f(1) = 0, f'(1) = 1.$

【解1】 $\frac{\partial z}{\partial x} = yf(\frac{y}{x}) - \frac{y^2}{x}f'(\frac{y}{x})$

$$\frac{\partial z}{\partial y} = xf(\frac{y}{x}) + yf'(\frac{y}{x})$$

$$x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = 2xyf(\frac{y}{x}) = xy \ln \frac{y}{x} \quad f(x) = \frac{1}{2} \ln x$$

【解2】

题型四 隐函数的偏导数与全微

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【例1】设 $z = z(x, y)$ 是由方程 $z + e^z = xy$ 所确定, 求 $\frac{\partial z}{\partial x}$ 和 $\frac{\partial z}{\partial y}$.

【解1】公式法 $\frac{\partial z}{\partial x} = -\frac{F_x}{F_z},$

$$\frac{\partial z}{\partial y} = -\frac{F_y}{F_z};$$

【解2】等式两边求导

【例3】利用微分形式不变性

【例2】设方程 $F(\frac{x}{z}, \frac{z}{y}) = 0$ 可确定函数 $z = z(x, y)$, 求 $\frac{\partial z}{\partial x}$ 和 $\frac{\partial z}{\partial y}$.

$$\frac{\partial z}{\partial x} = -\frac{\frac{1}{z}F_1}{F_1 \cdot (-\frac{x}{z^2}) + F_2 \cdot (-\frac{1}{y})}$$

$$(1 + e^z) \frac{\partial z}{\partial x} = y$$

【例4】设 $u = f(x, y, z)$, $\varphi(x^2, e^y, z) = 0$, $y = \sin x$ 确定了函数

$u = u(x)$, 其中 f, φ 都有一阶连续偏导数, 且 $\frac{\partial \varphi}{\partial z} \neq 0$, 求 $\frac{du}{dx}$.

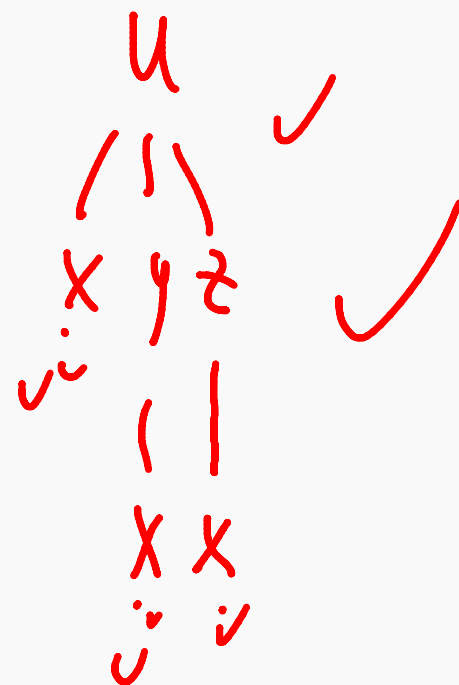
【解1】 $\frac{du}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \cos x + \frac{\partial f}{\partial z} \frac{dz}{dx}$

$\varphi(x^2, e^y, z) = 0$ 两端对 x 求导得

$$\varphi_1 2x + \varphi_2 e^y \cos x + \varphi_3 \frac{dz}{dx} = 0$$

解得 $\frac{dz}{dx} = -\frac{1}{\varphi_3} (2x\varphi_1 + \varphi_2 e^y \cos x)$

$$\frac{du}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \cos x - \frac{\partial f}{\partial z} (2x\varphi_1 + \varphi_2 e^y \cos x)$$



【例4】设 $u = f(x, y, z)$, $\varphi(x^2, e^y, z) = 0$, $y = \sin x$ 确定了函数

$u = u(x)$, 其中 f, φ 都有一阶连续偏导数, 且 $\frac{\partial \varphi}{\partial z} \neq 0$, 求 $\frac{du}{dx}$.

【解2】 $u = f(x, y, z)$

* $\varphi(x^2, e^y, z) = 0$

$y = \sin x$

$$dz = -\frac{1}{\varphi_3}(\varphi_1 2x + \varphi_2 e^y \cos x)dx$$

$$du = \left[\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \cos x - \frac{\partial f}{\partial z} (2x\varphi_1 + \varphi_2 e^y \cos x) \right] dx$$

$$du = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz$$

$$\varphi_1 2x dx + \varphi_2 e^y dy + \varphi_3 dz = 0$$

$$dy = \cos x dx$$

【例5】设 $y = f(x, t)$ 且方程 $F(x, y, t) = 0$ 确定了函数

$t = t(x, y)$, 求 $\frac{dy}{dx}$.

【解1】由 $y = f(x, t)$ 得

$$\frac{dy}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial t} \left(\frac{\partial t}{\partial x} + \frac{\partial t}{\partial y} \frac{dy}{dx} \right)$$

由 $F(x, y, t) = 0$ 得 $\frac{\partial t}{\partial x} = -\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial t}}, \quad \frac{\partial t}{\partial y} = -\frac{\frac{\partial F}{\partial y}}{\frac{\partial F}{\partial t}}$

$$\frac{dy}{dx} = \frac{\partial f}{\partial x} - \frac{\partial f}{\partial t} \left(\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial t}} + \frac{\frac{\partial F}{\partial y}}{\frac{\partial F}{\partial t}} \frac{dy}{dx} \right) \quad \frac{dy}{dx} = \frac{\frac{\partial F}{\partial t} \frac{\partial f}{\partial x} - \frac{\partial F}{\partial x} \frac{\partial f}{\partial t}}{\frac{\partial F}{\partial t} + \frac{\partial F}{\partial y} \frac{\partial f}{\partial t}}$$

【例5】设 $y = f(x, t)$, 且方程 $F(x, y, t) = 0$ 确定了函数

$t = t(x, y)$, 求 $\frac{dy}{dx}$.

【解2】 $y = f(x, t)$,

*

$$F(x, y, t) = 0$$

$$dy = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial t} dt$$

$$\frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy + \frac{\partial F}{\partial t} dt = 0$$

$$dt = -\frac{1}{\frac{\partial F}{\partial t}} \left(\frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy \right)$$

$$\frac{dy}{dx} = \frac{\frac{\partial F}{\partial t} \frac{\partial f}{\partial x} - \frac{\partial F}{\partial x} \frac{\partial f}{\partial t}}{\frac{\partial F}{\partial t} + \frac{\partial F}{\partial y} \frac{\partial f}{\partial t}}$$



还不关注，
你就慢了



$(y=y(x)) \text{ 直线} \Leftrightarrow y=ax+b \Leftrightarrow y''=0$
 $y'=a$
 $y=ax+b$

【例6】设 $f(x, y)$ 有二阶连续偏导数, 且 $f'_{xy} \neq 0$, 证明: 对任意

常数 C , $f(x, y) = C$ 为一条直线 $\Leftrightarrow f_2^2 f_{11} - 2f_1 f_2 f_{12} + f_1^2 f_{22} = 0$.

【证】

$f_1 + f_2 \frac{dy}{dx} = 0$
 $\frac{dy}{dx} = -\frac{f_1(x, y)}{f_2(x, y)}$

$\frac{d^2 y}{dx^2} = -\frac{d}{dx} \left(\frac{f_1}{f_2} \right)$
 $= -\frac{(f_{11} + f_{12} \frac{dy}{dx}) f_2 - (f_{21} + f_{22} \frac{dy}{dx}) f_1}{f_2^2}$
 $= -\frac{f_2^2 f_{11} - 2f_1 f_2 f_{12} + f_1^2 f_{22}}{f_2^3} = 0$

$y = y(x)$

$y = ax + b$

f'



还不关注，你就慢了



祝同学们

考研路上一路顺利!

$$u = f(x, y, z)$$

$$xe^x - ye^y = ze^z$$

$$\Rightarrow z = z(x, y)$$

$$\frac{\partial u}{\partial x} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial z} \frac{\partial z}{\partial x}$$

$$u = f(x, y, z)$$